

Algorithms

0-12
 O, Ω, Θ

Asymptotic notations

Lec_13

Introduction to Small o and Small Omega

Small o $\Rightarrow o$
Small omega $\Rightarrow \omega$

Big
 O
 Ω

Recap

$f(n), g(n)$

O

$$\begin{aligned} f(n) &\leq c \cdot g(n) \\ \Downarrow \Uparrow \\ f(n) &= O(g(n)) \end{aligned}$$

where

$$\exists n \geq n_0$$

$c = \text{constant}$

Ω

$$\begin{aligned} f(n) &\geq c \cdot g(n) \\ \Downarrow \Uparrow \\ f(n) &= \Omega(g(n)) \end{aligned}$$

where

Θ

$$\begin{aligned} f(n) &\leq c_1 g(n) \\ \text{and} \\ f(n) &\geq c_2 g(n) \\ \Downarrow \\ f(n) &= \Theta(g(n)) \\ \Downarrow \\ g(n) &= \Theta(f(n)) \end{aligned}$$

★ $a = O(b)$
 $\Downarrow$
 $a \leq c \cdot b$

★ $n = O(1)$
 $n \geq c \cdot 1$

★ $= \theta(1)$
 $p \leq c \cdot 1$
 and
 $p \geq c \cdot 1$

Small o

Big O $\Rightarrow$

$$f(n) \leq c \cdot g(n)$$

$\Downarrow$

$$f(n) = O(g(n))$$

$$\checkmark a < c \cdot b$$

$$\checkmark a = c \cdot b$$

$$a = O(b)$$

$\Downarrow$

$$a \leq c \cdot b$$

Ex:

$$f(n) < c \cdot g(n)$$

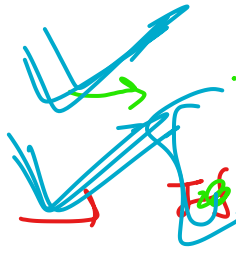
$\Downarrow$

$$f(n) = o(g(n))$$

$$\left[\begin{array}{l} n \leq y \\ n < y \\ \text{or} \\ n = y \end{array} \right]$$

$$\boxed{ \begin{array}{l} a = o(b) \\ \Downarrow \\ a < c \cdot b \end{array} }$$

True



If small o exist then Big O also exist

If Big O exist then small o also exist

False

Examples: ~~check~~ $f(n) = o(g(n))$, or $f(n) = O(g(n))$

1) $f(n) = n^2$, $g(n) = n^3$

$$\begin{array}{c|c} f(n) & g(n) \\ n^2 & n^3 \end{array}$$

$$f(n) < c \cdot g(n)$$

$$n^2 < c \cdot n^3$$

$\frac{f(n)}{g(n)}$

$$f(n) < c \cdot g(n)$$

$$n^2 \leq c \cdot n^3$$

$$f(n) < c \cdot g(n)$$

$$f(n) = o(g(n))$$

$$f(n) = O(g(n))$$

2) $f(n) = n^3$

$$f(n) < c \cdot g(n)$$

$$n^3 < n^3 + 2$$

$$g(n) = n^3 + 2$$

$$n^3 < c(n^3 + 2)$$

$$n^3 = c \cdot (n^3 + 2)$$

$$n^3 = n^3 + \cancel{2c}$$

$$f(n) < c \cdot g(n)$$

Big O

$$f(n) = O(g(n))$$

→ If highest power for both fn is same, small o never exist

Small omega:

Ω

$$f(n) \geq c \cdot g(n)$$

$$\Downarrow$$
$$f(n) = \Omega(g(n))$$

w
$$f(n) > c \cdot g(n)$$
$$\Downarrow$$
$$f(n) = \omega(g(n))$$

Ex:

$$f(n) = n^6$$

$$g(n) = n^5$$

$$\checkmark \begin{array}{l} f(n) > g(n) \\ n^6 > c \cdot n^5 \end{array}$$

$$\begin{array}{l} f(n) = c \cdot g(n) \\ n^6 = c \cdot n^5 \end{array}$$

$$\text{if } f(n) > c \cdot g(n)$$
$$f(n) = \omega(g(n))$$

$\Rightarrow$ In small o and small ω equal condition is excluded from the Big O and Big Ω

$$\underline{f(n)} \geq (g(n))$$

Examples:

$\hookrightarrow O, \Omega, \omega$

1) $\checkmark n^3 = O(n^5)$
 $\checkmark n^5 = O(n^4)$
 $\checkmark n^3 = O(n^3)$
 ~~$n^3 = O(n^2)$~~

$$f(n) = O(g(n))$$

$$f(n) \ll g(n)$$

~~$n^3 = O(n^5)$~~
 ~~$n^3 = O(n^4)$~~
 $n^3 = O(n^3)$
 $n^3 = O(n^2)$
 ~~$n^3 \triangleq n^2$~~

$\checkmark n^6 = \Omega(n^3)$
 $\checkmark n^6 = \Omega(n^4)$
 $\checkmark n^6 = \Omega(n^5)$
 $\checkmark n^6 = \Omega(n^6)$
 ~~$n^6 = \Omega(n^7)$~~
 ~~$n^6 = \Omega(n^8)$~~

$\checkmark n^6 = \omega(n^3)$
 $\checkmark n^6 = \omega(n^4)$
 $\checkmark n^6 = \omega(n^5)$
 ~~$n^6 = \omega(n^6)$~~
 ~~$n^6 = \omega(n^7)$~~
 ~~$n^6 = \omega(n^8)$~~

⇒ equal condition

Theta Notation imp point: power of highest term is same

Example:

$$f(n) \leq c_1 g(n) \\ \text{and} \\ f(n) \geq c_2 g(n)$$

Ex:

$$f(n) = n^2 + n + 2 \\ g(n) = 6n^2 + 6 + n$$

$$f(n) \leq 4 g(n)$$

$$\textcircled{n^2} + n + 2 \leq \textcircled{41} (\textcircled{6n^2} + 6 + n) \\ \hookrightarrow \textcircled{no}$$

if highest power
of both fn is
same
⇒
equal
⇒
theta condⁿ

⇒ Theta notation
equal condition

{small o, small w} ⇒ equal condition is excluded

→ If theta is there, means.
Equal condition is there and we
know that if equal is there
small o and small w can not exist

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